Lecture 11: Capacitance

Pre-Lecture Worksheet

Complete the following worksheet in preparation for the next lecture. The following questions correspond to material covered in chapter 24. Feel free to read chapter 24 and answer the questions. You may also seek answers to the questions from other sources. Note: the textbook will be used as the final source for these questions. While answers gleaned from other sources may be correct, answers must encompass the material covered in the book to be considered correct.

1) If you want your system to do more work, you need to increase the capacitance, C, of the system. For the following systems:

(a) label the charges on the battery

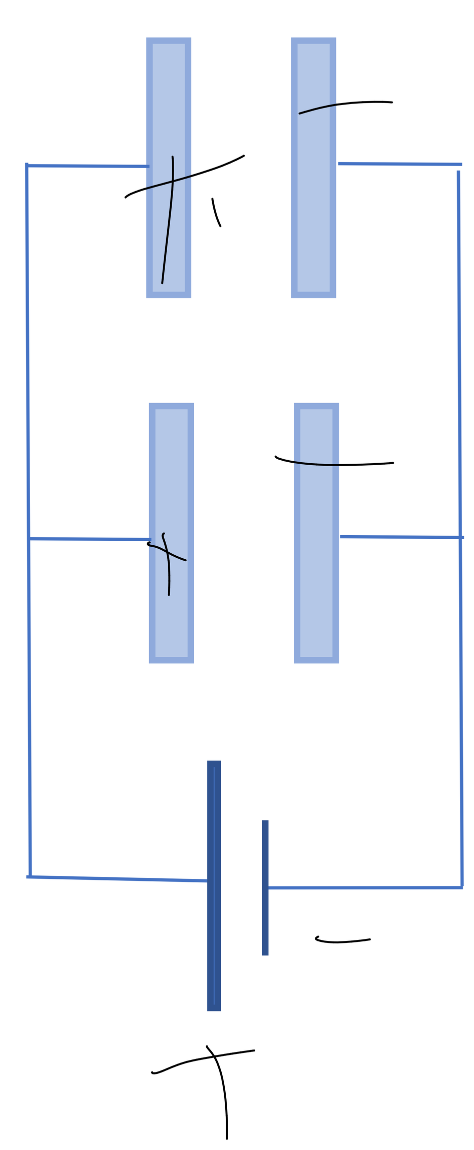
(b) label the capacitors with the appropriate charges

(c) label whether the system is parallel or series

(d) write the appropriate equation for the equivalent capacitance of the system

(e) state whether the system adds capacitance or voltage.

A

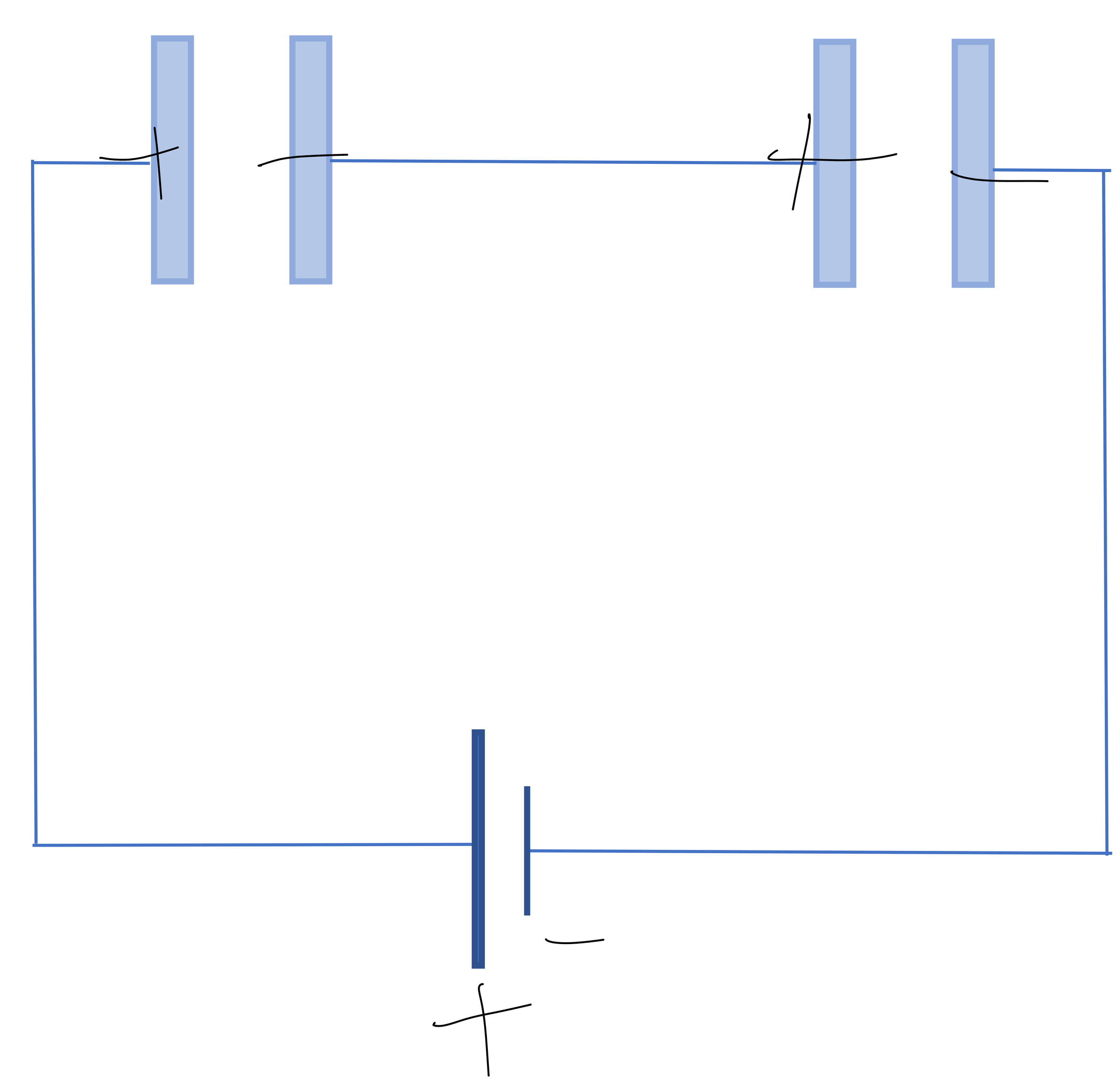


Parallel

$C = C_1 + C_2$

Capacitance

B



Series

$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$

voltage

2) (True or false) dielectrics increase the electric field.

False

3) Change the following equation to make the equation for the dielectric constant, K, true.

$$C_{m} = K*C_{o}$$